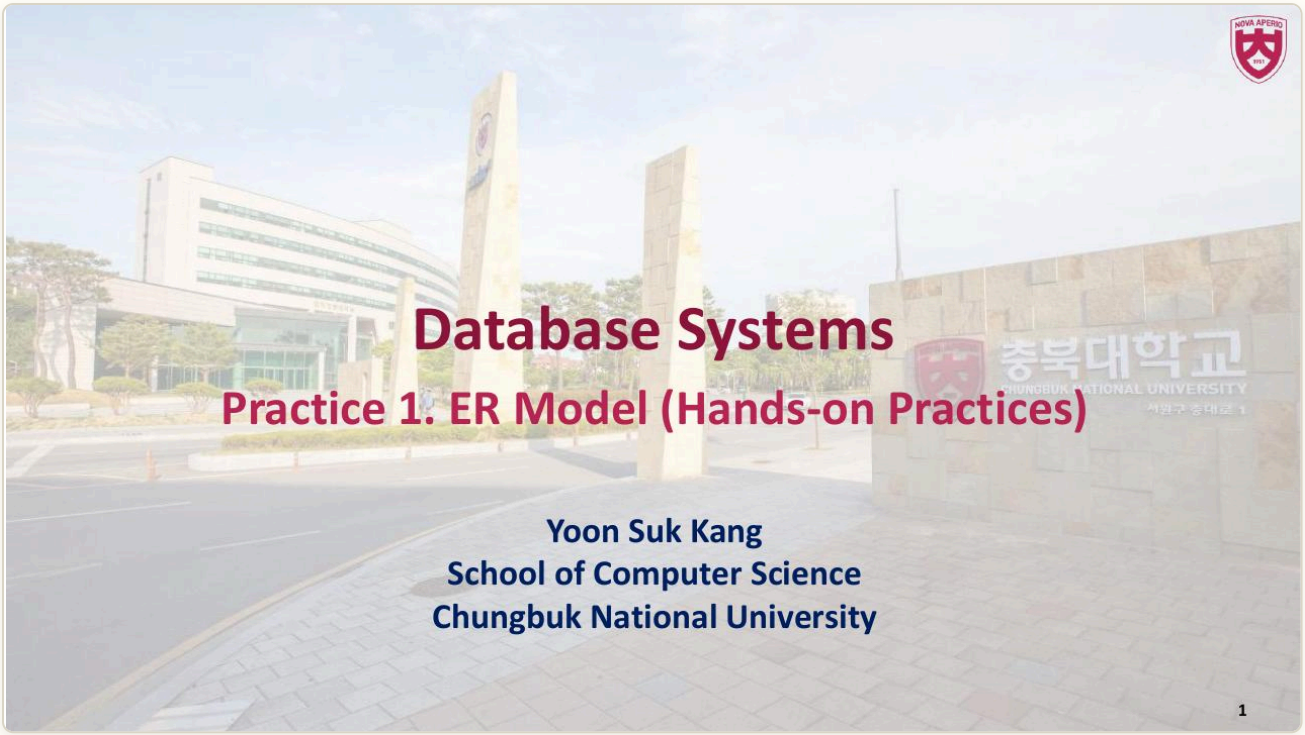
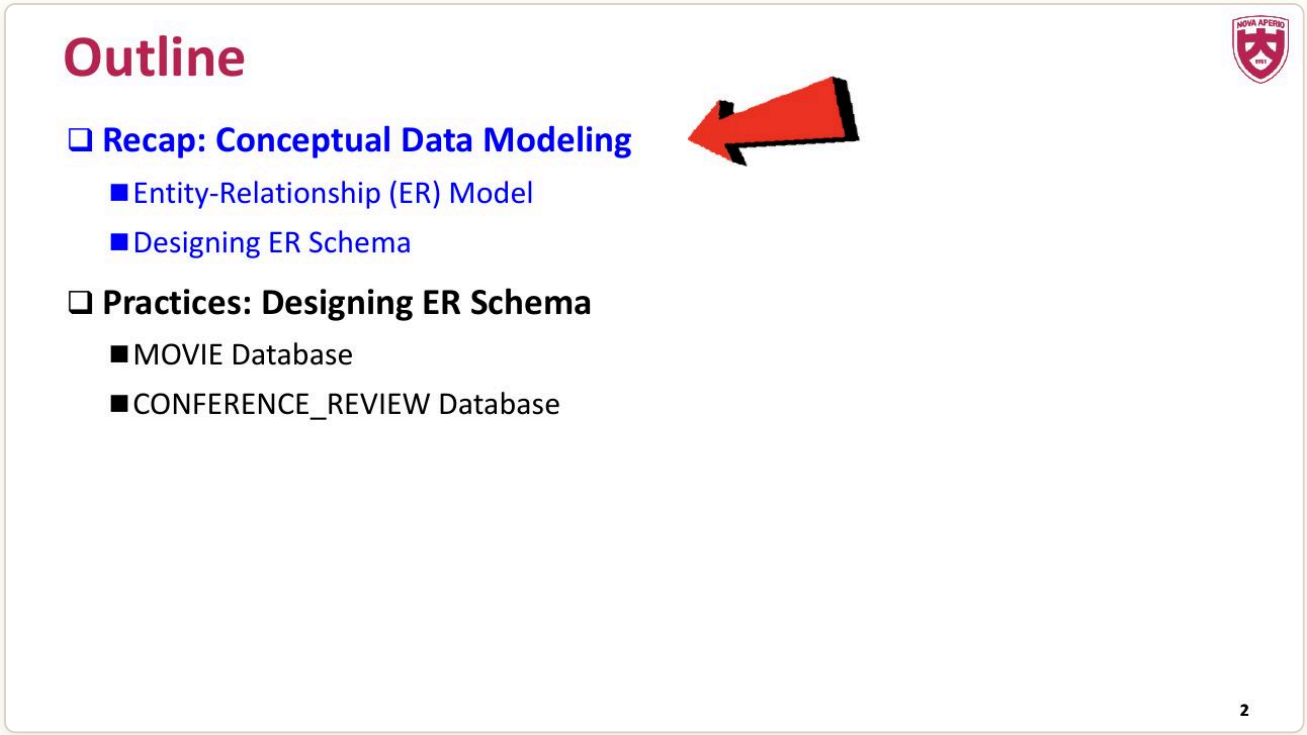


데이터베이스 시스템 — 실습 1: ER 모델

- Yoon Suk Kang · 충북대 소프트웨어학부

필기 ✍️

💡 이해 요구사항 문장 ⟶ ER 다이어그램 연습. 시험 서술형(ER 그리기) 대비 1순위



목차 (Outline)

- **복습: 개념적 데이터 모델링** — ER 모델 / ER 스키마 설계
- **실습: ER 스키마 설계** — MOVIE DB / CONFERENCE_REVIEW DB

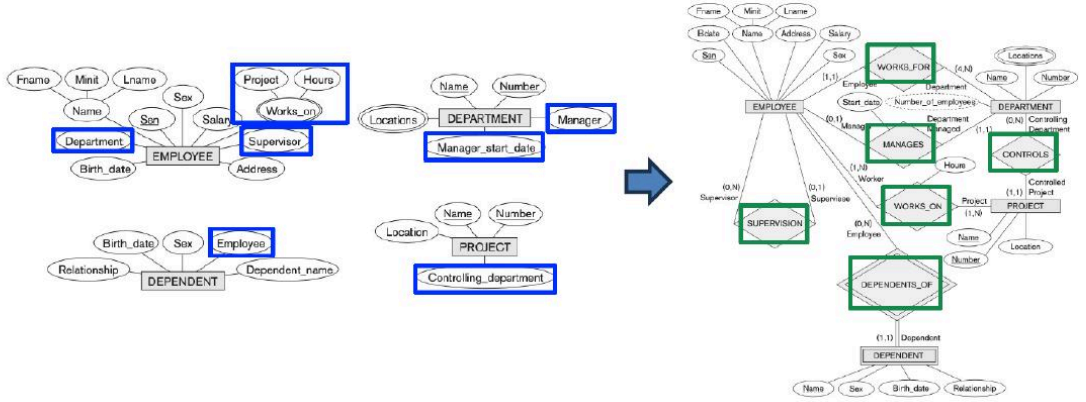
필기 ✍️

— 넘어가도 되는 쪽

Summary: Designing ER Schema

❑ Process of Designing ER Schema Diagram

- ■(1) Describe entity types and their attributes based on the data requirements
- ■(2) **Specify relationship types and their constraints**
- ■(3) **Change the attributes representing relationships into relationship types**



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요약: ER 스키마 설계

- ER 스키마 다이어그램 설계 과정
  - (1) 요구사항에 따라 **엔티티 타입과 속성** 기술
  - (2) **관계 타입과 제약** 지정
  - (3) 관계를 나타내던 **속성을 관계 타입으로** 바꾸기

필기 ✍️

★ 시험 아래 두 실습 모두 이 Step 1 ⟶ 2 ⟶ 3 순서

Recap: Constrains on Relationship Types



❑ **Constraints of Relationship Types**

- ■ Relationship types usually have certain constraints about participating entities
  - ❑ E.g., the company has a rule that each employee must work for exactly one department

❑ **Constraint 1: Cardinality Ratio**

- ■ Specify the maximum number of relationship instances that an entity can participate in
- ■ Three types of cardinality ratios for a binary relationship
  - ❑ 1:1, 1:N, M:N

❑ **Constraint 2: Participation Constraint**

- ■ Specifies whether the existence of an entity depends on its being related to another entity
- ■ Two types of a participation constraint
  - ❑ **Total**: every entity should participate in the relationship
  - ❑ **Partial**: every entity does not need to participate in the relationship

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복습: 관계 타입의 제약

- 관계 타입에는 보통 참여 엔티티에 대한 제약이 있다
  - 예: 회사 규칙 — 직원은 정확히 한 부서에서 일한다
- **제약 1: Cardinality Ratio** — 엔티티가 참여할 수 있는 관계 인스턴스의 **최대** 수
  - 이항 관계: 1:1, 1:N, M:N
- **제약 2: Participation Constraint** — 엔티티의 존재가 다른 엔티티와의 관계에 **의존**하는지
  - **Total**: 모든 엔티티가 참여 / **Partial**: 모두 참여할 필요 없음

필기 ✍️

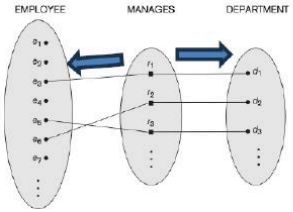
— 넘어가도 되는 쪽

Recap: Attributes of Relationship Types

❑ Relationship Types Can Have Attributes

The company wants to keep track of:
(1) the **start date** of a manager when the employee manage the department;
(2) the **start date** of an employee when the employee works for the department
(3) the **number of working hours** of each employee on a specific project and

1:1 Relationship Type

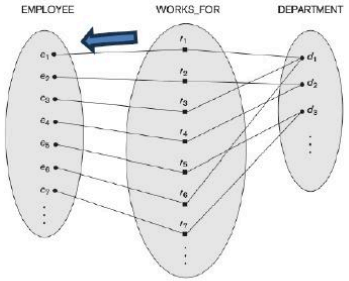


StartDate for MANAGES

Migrated to **one of any participating entities**

Which one is better?

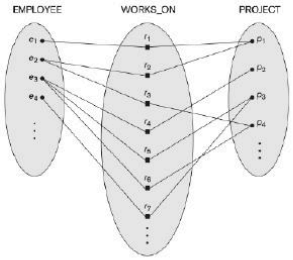
1:N Relationship Type



StartDate for WORKS_FOR

Migrated only to **the N-side participating entity**

M:N Relationship Type



Hours for WORKS_ON

Cannot be migrated to any entities

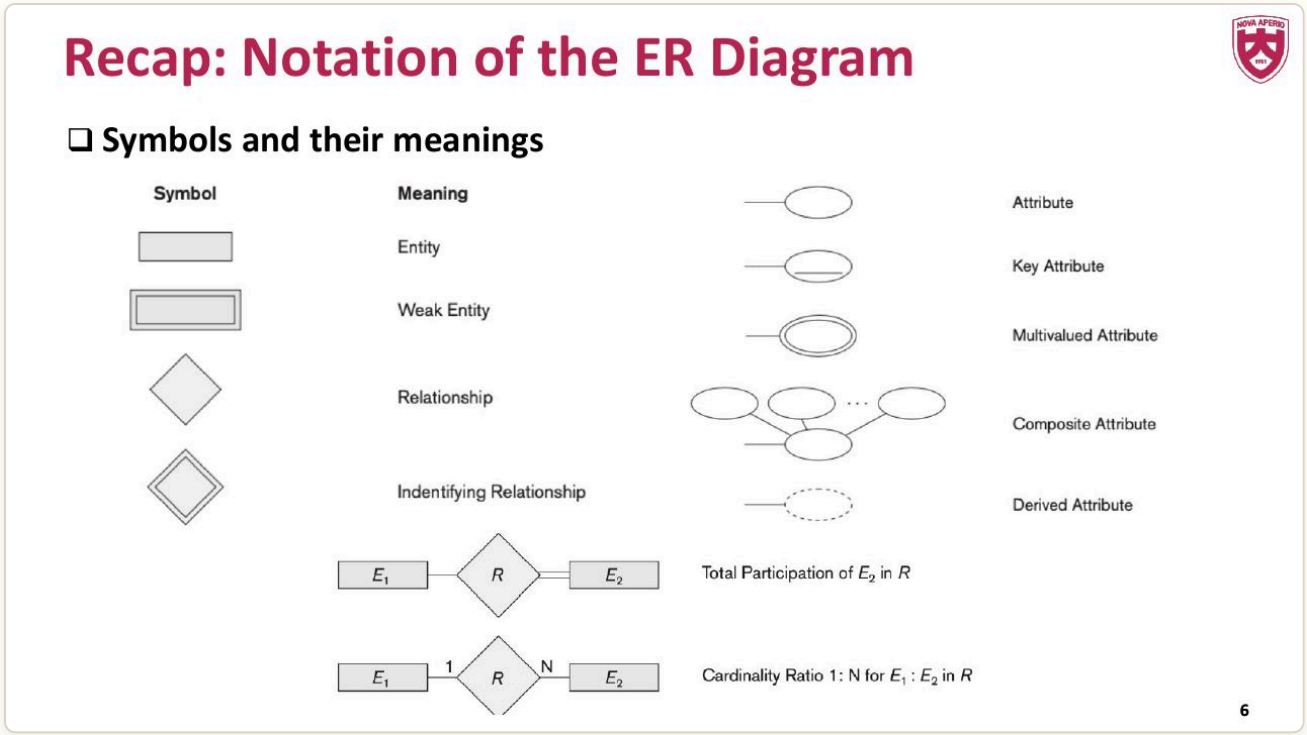
5

복습: 관계 타입의 속성

- 관계 타입도 속성을 가질 수 있다
  - MANAGES의 StartDate, WORKS_ON의 Hours

필기 ✍️

★ 시험 M:N 관계의 속성은 관계에 붙는다 — 이번 실습의 Role·Quotes·Rating이 그 예

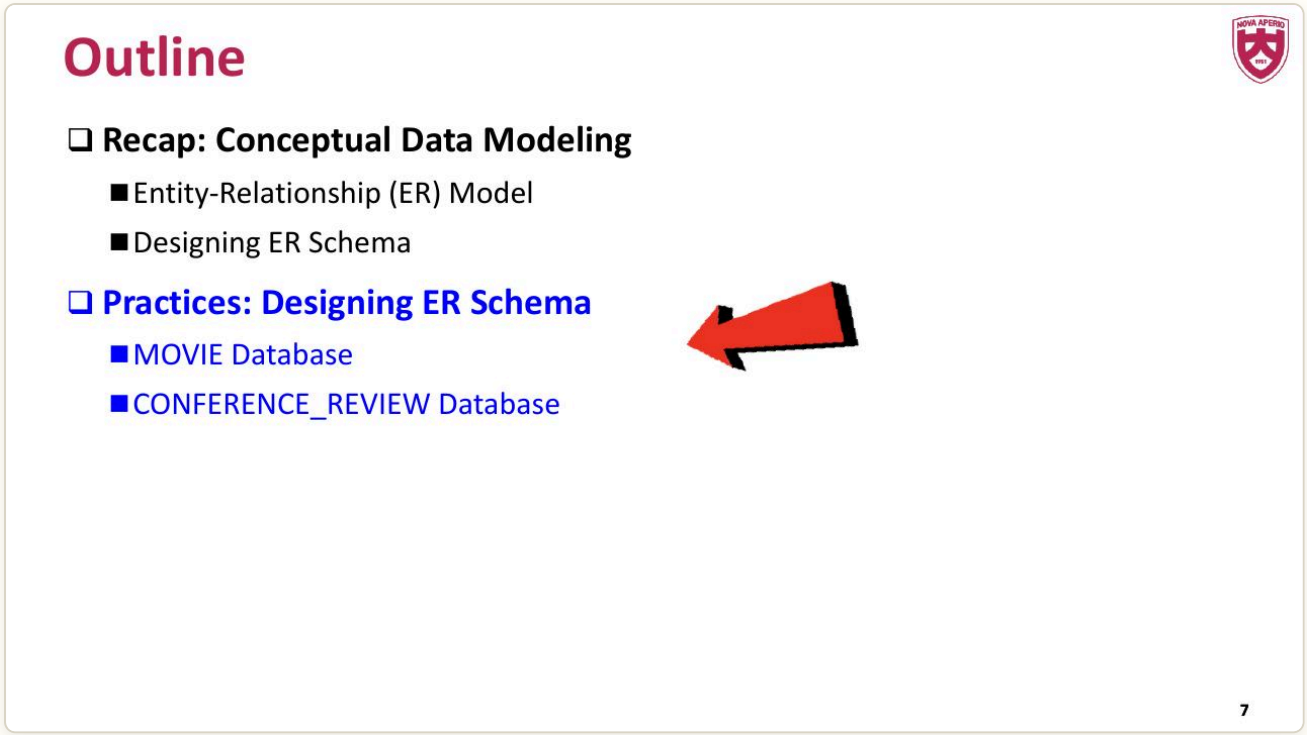


복습: ER 다이어그램 기호

- 사각형 = 엔티티 / 이중 사각형 = 약한 엔티티
- 마름모 = 관계 / 이중 마름모 = 식별 관계
- 타원 = 속성 / 밑줄 = 키 / 이중 타원 = 다중값 / 타원에 타원 = 복합 / 점선 타원 = 유도
- 이중선 = 전체 참여 / 1, N = 카디널리티 / (min, max) = 구조적 제약

필기 ✍️

★ 시험 기호표는 2강 P.50·P.70과 같다



목차 (Outline)

- **복습: 개념적 데이터 모델링** — ER 모델 / ER 스키마 설계
- **실습: ER 스키마 설계** — MOVIE DB / CONFERENCE_REVIEW DB

필기 ✍️

💡 이해 실습 1: MOVIE

Exercise 1: ER Schema Diagram

❑Designing ER Diagram for MOVIE Database

■Consider a MOVIE database in which data is recorded about the movie industry

■The data requirements are summarized as follows:



❑MOVIE

- ■ Each movie is identified by (**title and year of release**)
- ■ Each has a length in minutes and a plot outline
- ■ Each is classified under one or more genres (e.g., horror, action, drama, ...)
- ■ Each movie has a production company, one or more directors, and one or more actors appearing in it



❑ACTOR

- ■ Actors are identified by (**name and date of birth**) and appear in one or more movies
- ■ Each actor has a role in the movie and has zero or more quotable quotes to speak





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연습 1: MOVIE DB ① 요구사항

- 영화 산업 데이터를 기록하는 MOVIE DB
- **MOVIE**
  - (제목, 개봉 연도)로 식별
  - 상영 시간(분)과 줄거리 요약이 있음
  - 장르 **하나 이상**으로 분류 (공포, 액션, 드라마 ...)
  - 제작사 하나, 감독 **한 명 이상**, 배우 **한 명 이상**
- **ACTOR**
  - (이름, 생년월일)로 식별, 영화 **하나 이상**에 출연
  - 영화마다 배역(role)이 있고, 명대사(quotes)가 **0개 이상**

필기 ✍️

💡 이해 '(title and year)로 식별' → 복합 키 / 'one or more genres' → 다중값

Exercise 1: ER Schema Diagram (cont.)

❑Designing ER Diagram for MOVIE Database

■Consider a MOVIE database in which data is recorded about the movie industry

■The data requirements are summarized as follows:



❑**DIRECTOR**

■ Directors are identified by (**name and date of birth**) and direct one or more movies

■ It is possible for a director to act in a movie (including one that she/he may also direct)



❑**PRODUCTION**

■ Production companies are identified by (**name**) and each has an address

■ A production company produces one or more movies





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연습 1: MOVIE DB ② 요구사항 (계속)

- **DIRECTOR**
  - (이름, 생년월일)로 식별, 영화 **하나 이상**을 감독
  - 감독이 영화에 출연할 수도 있음 (자기가 감독한 영화 포함)
- **PRODUCTION**
  - 제작사는 (이름)으로 식별, 주소가 있음
  - 제작사는 영화를 **하나 이상** 제작

필기 ✍️

⚠ 주의 감독도 출연 가능 → 감독-영화 사이 '출연' 관계가 따로 필요 (DIR_ACTS)

Step 1: Describing the Entity Types



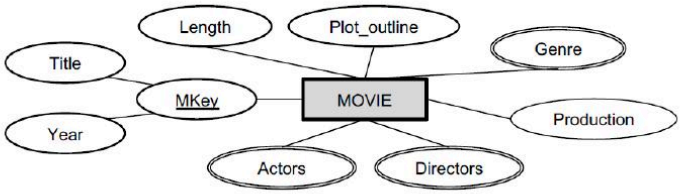
❑ **Entity Type 1: MOVIE**

■ Requirements:

- ❑ Each movie is identified by (**title and year of release**)
- ❑ Each has a length in minutes and a plot outline
- ❑ Each is classified under one or more genres (e.g., horror, action, drama, ...)
- ❑ Each movie has a production company, one or more directors, and one or more actors appearing in it

■ Attributes:

- ❑ *Mkey(Title, Year), Length, Plot_outline, {Genre}, Production, {Directors}, {Actors}*



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Step 1: 엔티티 타입 — MOVIE

- 요구사항: (제목, 연도)로 식별 / 상영 시간, 줄거리 / 장르 하나 이상 / 제작사, 감독들, 배우들
- 속성:

```
Mkey(Title, Year), Length, Plot_outline, {Genre}, Production, {Directors}, {Actors}
```

필기 ✍️

★ 시험 { } = 다중값, ( ) = 복합. Mkey는 복합 키

💡 이해 Production·Directors·Actors는 다른 엔티티를 가리키는 속성 ⟶ Step 2에서 관계로 바뀜

Step 1: Describing the Entity Types (cont.)

❑ **Entity Type 2: ACTOR**

■ Requirements:

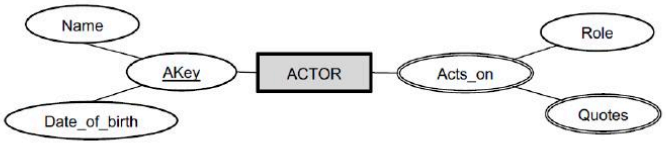
❑ Actors are identified by (**name and date of birth**) and appear in one or more movies

❑ Each actor has a role in the movie and has zero or more quotable quotes to speak



■ Attributes:

❑ *Akey(Name, Date_of_birth), {Acts_on(Movie, Role, {Quotes})}*



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Step 1: 엔티티 타입 — ACTOR

- 요구사항: (이름, 생일)로 식별, 영화 하나 이상 출연 / 배역과 명대사 0개 이상
- 속성:

```
Akey(Name, Date_of_birth), {Acts_on(Movie, Role, {Quotes})}
```

필기 ✍️

✏ 예.풀이 Acts_on = 다중값 + 복합, 그 안에 Quotes도 다중값

Step 1: Describing the Entity Types (cont.)

❑ **Entity Type 3: DIRECTOR**

■ Requirements:

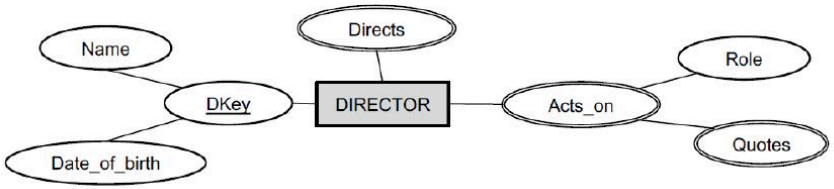
❑ Directors are identified by (**name and date of birth**) and direct one or more movies

❑ It is possible for a director to act in a movie (including one that she/he may also direct)



■ Attributes

❑ *Dkey(Name, Date_of_birth), {Directs}, {Acts_on(Movie, Role, {Quotes})}*



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Step 1: 엔티티 타입 — DIRECTOR

- 요구사항: (이름, 생일)로 식별, 영화 하나 이상 감독 / 영화에 출연할 수도
- 속성:

```
Dkey(Name, Date_of_birth), {Directs}, {Acts_on(Movie, Role, {Quotes})}
```

필기 ✍️

— 넘어가도 되는 쪽

Step 1: Describing the Entity Types (cont.)

❑ **Entity Type 4: PRODUCTION**

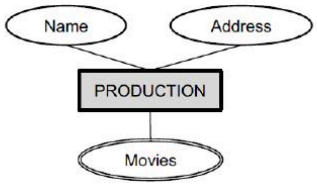
■ Requirements:

❑ Production companies are identified by (**name**) and each has an address

❑ A production company produces one or more movies

■ Attributes:

❑ *Name, Address, {Movies}*



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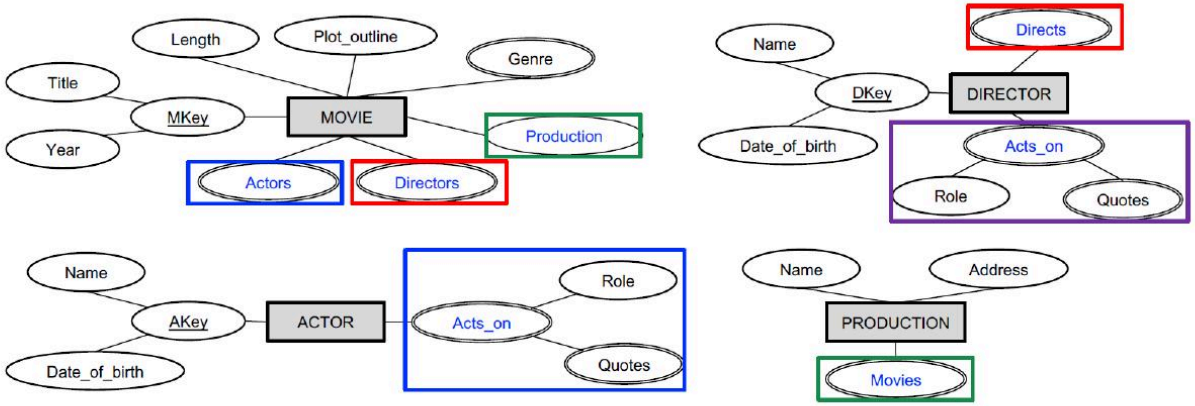
Step 1: 엔티티 타입 — PRODUCTION

- 요구사항: (이름)으로 식별, 주소 / 영화 하나 이상 제작
- 속성:

```
Name, Address, {Movies}
```

필기 ✍️

— 넘어가도 되는 쪽



Step 2: 관계 타입은 몇 개?

- **PRODUCES**, **ACTS**(배우 & 감독), **DIRECTS**

필기 ✍️

💡 이해 ACTS는 배우용(ACTS)과 감독용(DIR_ACTS) 두 개로 그려짐 (p.18)

Step 2: Describing the Relationship Types

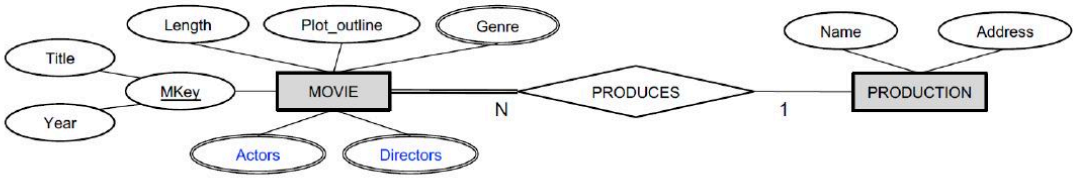
❑ **Relationship Type 1: PRODUCES**

■ Relating **PRODUCTION** and **MOVIE** entity types

❑ A single production company can produce many movies, but a movie has to be produced by a single production company

■ i.e., Cardinality ratio of this relationship type: (1:N)

❑ The MOVIE entity type has a total participation constraint



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관계 1: PRODUCES

- PRODUCTION과 MOVIE를 잇는다
  - 제작사 하나가 영화 여러 편을 제작, 영화는 제작사 **하나**가 제작
  - 카디널리티: **1:N**
  - MOVIE는 **전체 참여**

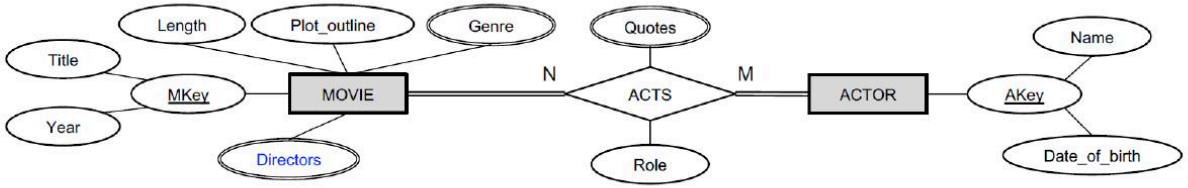
필기 ✍️

★ 시험 PRODUCES 1:N, MOVIE total

Step 2: Describing the Relationship Types (cont.)

❑ Relationship Type 2: ACTS

- ■ Relating **ACTOR** and **MOVIE** entity types
  - ❑ An actor can act in many movies, and a movie can have many actors
    - ■ i.e., Cardinality ratio of this relationship type: (M:N)
  - ❑ This relationship type has two relationship attributes *Role* and *Quotes* (the multivalued attribute)
    - ■ The *Role* attribute names a role that the actor plays in the movie, and the *Quotes* are quotes spoken by an actor
  - ❑ The both entity types have a total participation constraint



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관계 2: ACTS

- ACTOR와 MOVIE를 잇는다
  - 배우는 여러 영화에, 영화에는 여러 배우 — **M:N**
  - 관계 속성 2개: **Role**, **Quotes**(다중값)
    - Role = 배우가 그 영화에서 맡은 역, Quotes = 배우가 한 대사
  - **양쪽 모두 전체 참여**

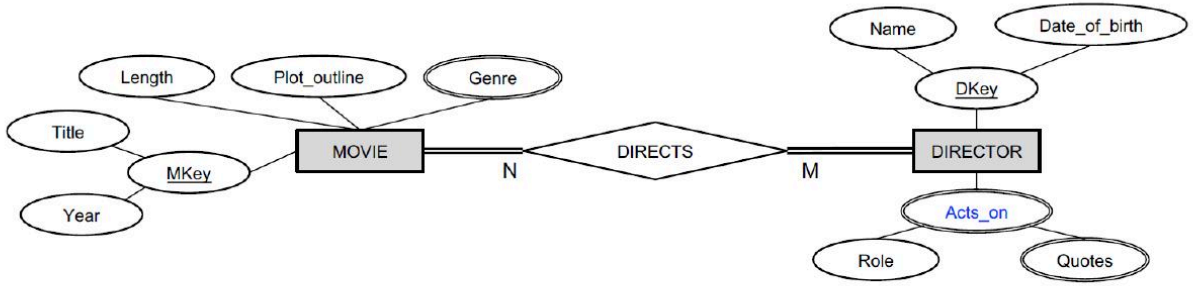
필기 ✍️

★ 시험 Role·Quotes는 배우-영화 조합으로 정해짐
→ 관계 속성 (M:N이라 엔티티로 못 올림)

Step 2: Describing the Relationship Types (cont.)

❑ Relationship Type 3: DIRECTS

- ■ Relating DIRECTOR and MOVIE entity types
  - ❑ A director can direct many movies, and a movie is directed by at least one director
    - ■ i.e., Cardinality ratio of this relationship type: (M:N)
  - ❑ The MOVIE entity type has a total participation constraint



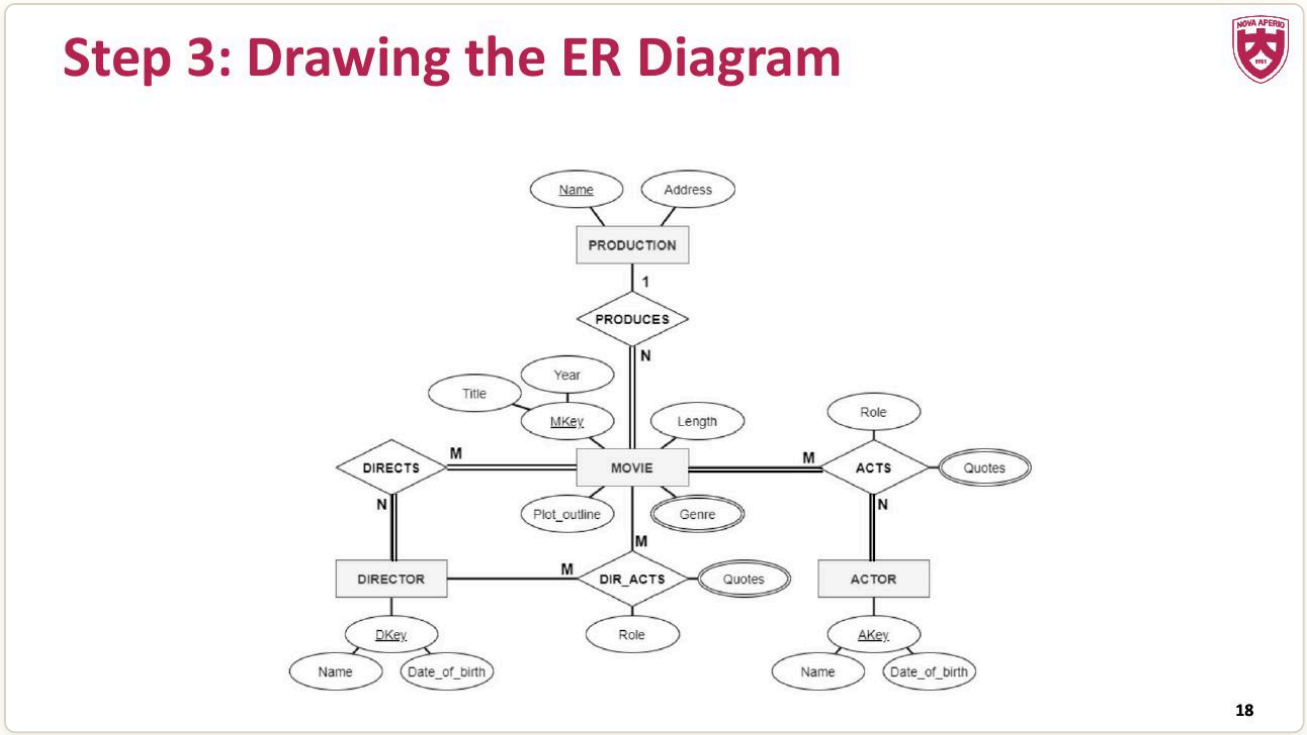
17

관계 3: DIRECTS

- DIRECTOR와 MOVIE를 잇는다
  - 감독은 여러 영화를 감독, 영화는 감독 **최소 한 명** — **M:N**
  - MOVIE는 **전체 참여**

필기 ✍️

⚠ 주의 그림(P.18)은 DIRECTOR 쪽도 이중선(감독은 영화 하나 이상)



Step 3: ER 다이어그램 (MOVIE)

- PRODUCTION(Name, Address) —1— PRODUCES —N═ MOVIE
- MOVIE(MKey(Title, Year), Length, Plot_outline, {Genre})
- MOVIE ═M— ACTS —N═ ACTOR, 관계 속성 Role, {Quotes}
- MOVIE ═M— DIRECTS —N═ DIRECTOR
- MOVIE —M— DIR_ACTS —M— DIRECTOR, 관계 속성 Role, {Quotes}
- ACTOR(AKey(Name, Date_of_birth)), DIRECTOR(DKey(Name, Date_of_birth))

필기 ✍️

★ 시험 정답 외우기: PRODUCES 1:N(MOVIE total) · ACTS M:N(Role, {Quotes}, 양쪽 total) · DIRECTS M:N(MOVIE total)

✏ 예.풀이 DIR_ACTS = 감독이 출연하는 경우 — 양쪽 partial

Exercise 2: Conference Review Database

❑ Background

- ■ Consider a CONFERENCE REVIEW database in which researchers submit their research papers for publication
- ■ Reviews by reviewers are recorded for use in the paper selection process
- ■ The database system caters primarily to reviewers who record answers to evaluation questions for each paper they review and make recommendations regarding whether to accept or reject the paper



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연습 2: CONFERENCE REVIEW DB — 배경

- 연구자들이 논문을 제출하는 학회 심사 DB
  - 심사위원의 심사가 기록되어 논문 선정에 쓰임
  - 주 사용자는 심사위원 — 논문마다 평가 질문에 답하고, 채택/거절 추천을 남김

필기 ✍️

— 넘어가도 되는 쪽

Data Requirements

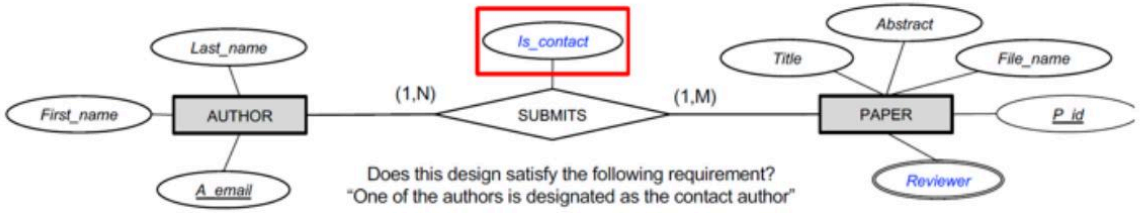
❑ **AUTHOR**

- Authors of papers are uniquely identified by **e-mail id** and have the first and last names
- Authors can submit one or more papers

❑ **PAPER**

- Each paper is assigned a unique id by the system and is described by a title, abstract, and the file

❑



- A reviewer rates each paper assigned to her/him on a scale of **1 to 10** in four evaluation categories: (1) technical merit, (2) readability, (3) originality, and (4) relevance to the conference
- Each reviewer provides an overall recommendation to each paper assigned to her/him
- Each review **contains two types of written comments**: (1) one to be seen by the review committee only and (2) the other as feedback to author(s)

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연습 2: 데이터 요구사항

- **AUTHOR**: 이메일로 유일하게 식별, 이름·성 / 논문을 하나 이상 제출
- **PAPER**: 시스템이 유일한 id 부여, 제목·초록·파일 이름
  - 저자 여러 명 가능, 그중 한 명이 **교신 저자(contact author)**
  - 논문 하나에 심사위원 **2~4명** 배정
- **REVIEWER**: 이메일로 유일하게 식별, 이름·성·전화·소속·관심 주제
  - 배정된 논문을 4개 항목에 1~10점으로 평가: (1) 기술적 가치 (2) 가독성 (3) 독창성 (4) 학회 관련성
  - 논문마다 종합 추천
  - 심사평 두 종류: (1) 심사위원회만 보는 것 (2) 저자에게 주는 피드백

필기 ✍️

💡 이해 숫자 표현('2~4명', '하나 이상')은 (min, max)로 바로 옮긴다

Step 1: Describing the Entity Types

❑ **Entity Type 1: AUTHOR**

■ Requirements:

❑ Authors of papers are uniquely identified by **e-mail id** and have the first and last names

❑ Authors can submit one or more papers

■ (Paper) Each can have multiple authors, but one of the authors is designated as the contact author

■ Attributes (List entity types and their attributes based on the requirements):



21

Step 1: AUTHOR (문제)

- 요구사항을 보고 AUTHOR의 속성을 적어라

필기 ✍

✏ 예.풀이 답은 p.29

Step 1: Describing the Entity Types (cont.)

❑ **Entity Type 2: PAPER**

■ Requirements:

- ❑ Each paper is assigned a **unique id** by the system and described by a title, abstract, and the file name
- ❑ Each can have multiple authors, but one of the authors is designated as the contact author
- ❑ A paper is assigned between two and four reviewers

■ Attributes (List entity types and their attributes based on the requirements):



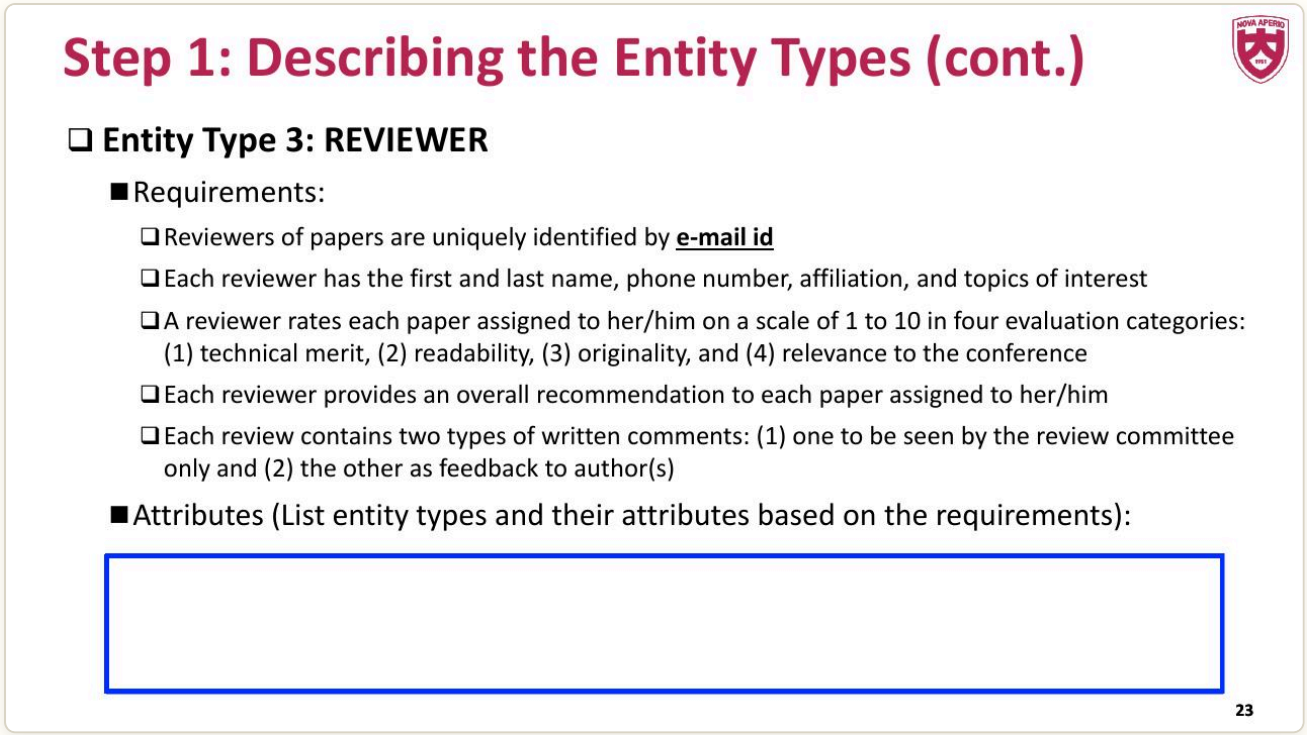
22

Step 1: PAPER (문제)

- 요구사항을 보고 PAPER의 속성을 적어라

필기 ✍️

✏ 예.풀이 답은 p.30

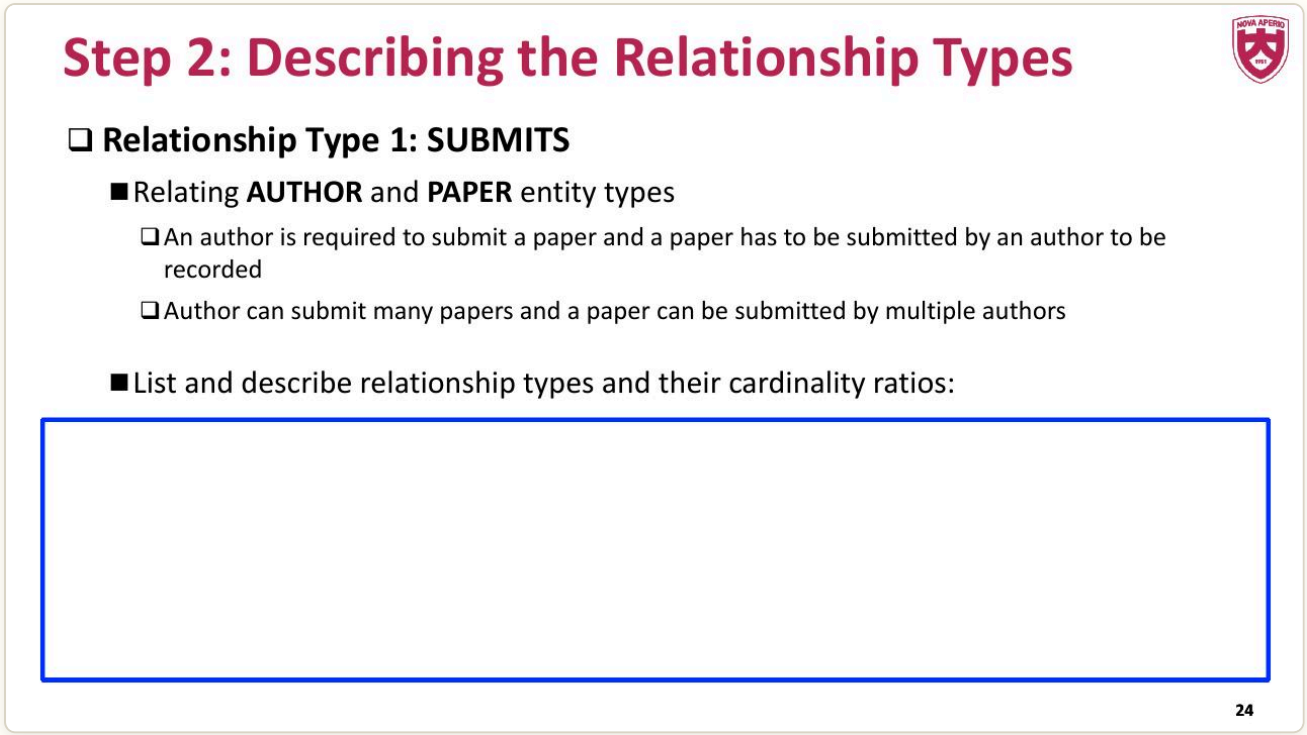


Step 1: REVIEWER (문제)

- 요구사항을 보고 REVIEWER의 속성을 적어라

필기 ✍️

✏ 예.풀이 답은 p.31~32

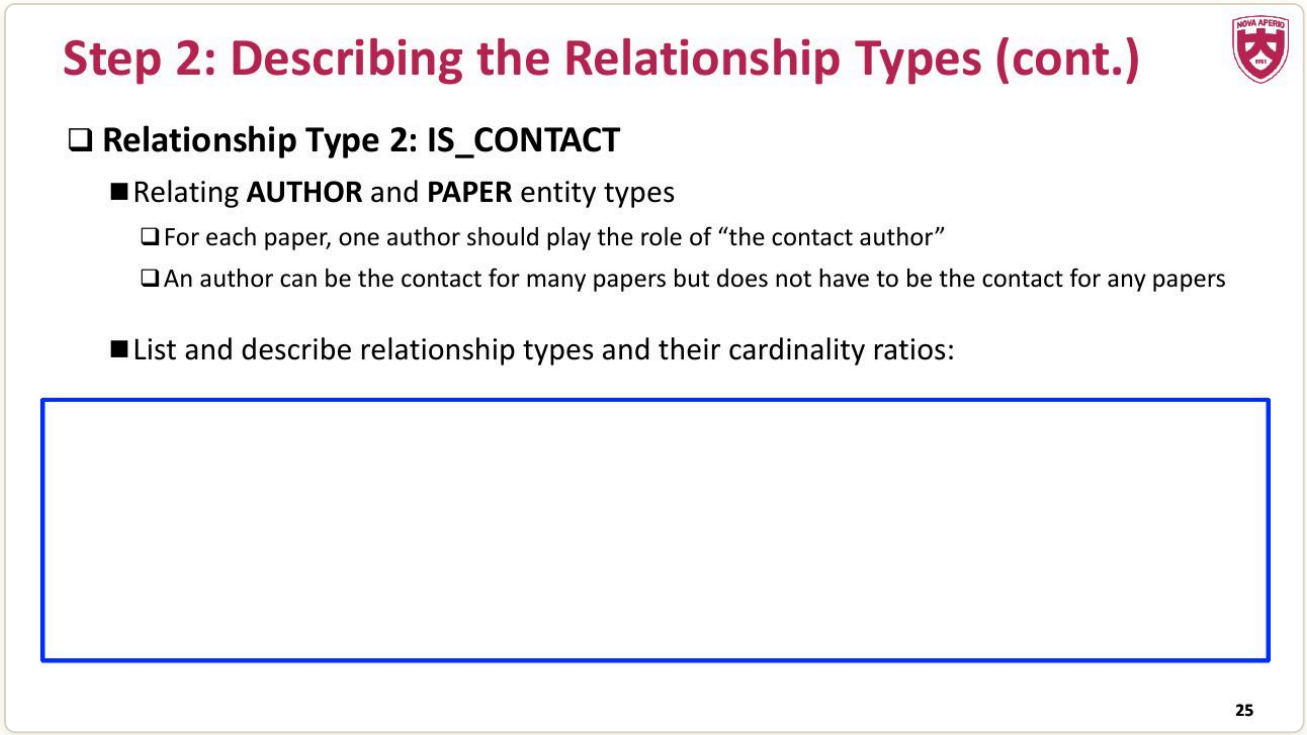


Step 2: SUBMITS (문제)

- AUTHOR와 PAPER를 잇는 관계
  - 저자는 논문을 제출해야 기록되고, 논문은 저자가 제출해야 기록된다
  - 저자는 여러 논문을, 논문은 여러 저자가 제출 가능

필기 ✍️

예.풀이 답은 P.33

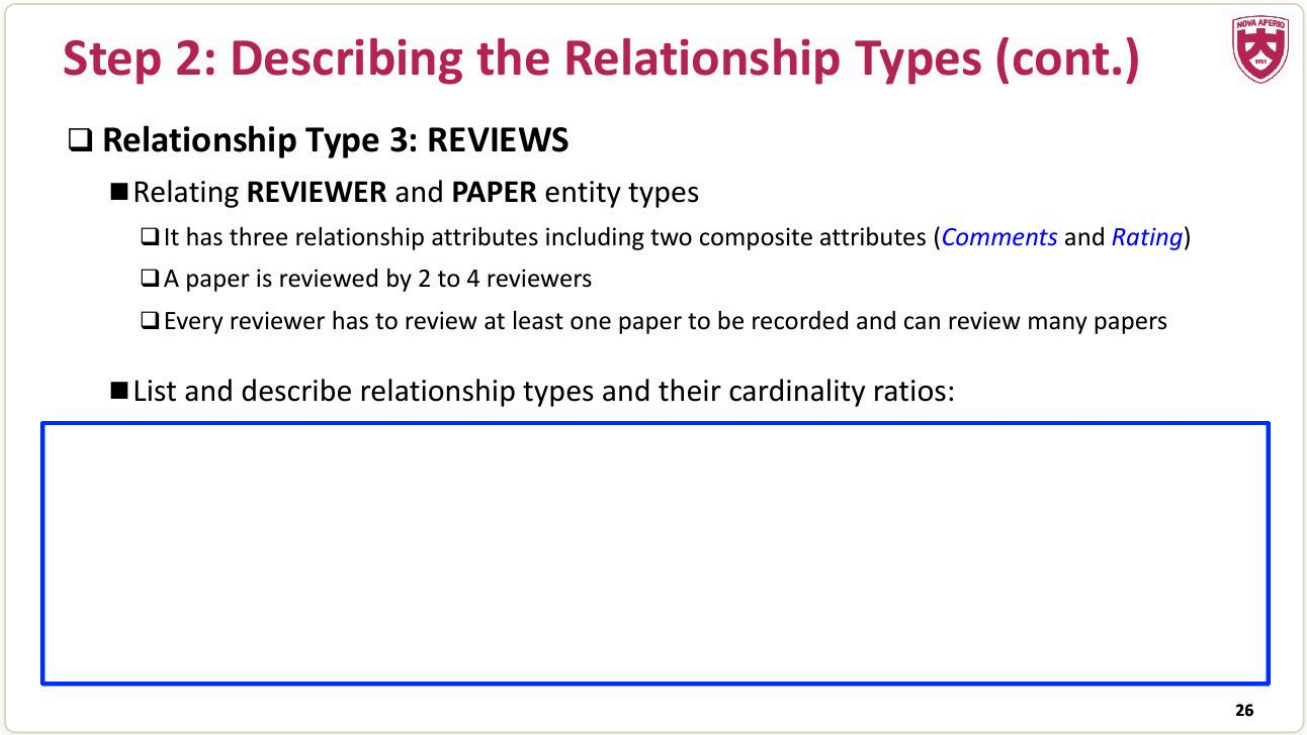


Step 2: IS_CONTACT (문제)

- AUTHOR와 PAPER를 잇는 관계
  - 논문마다 저자 한 명이 '교신 저자' 역할
  - 저자는 여러 논문의 교신 저자일 수 있지만, 아무 논문의 교신 저자가 아니어도 됨

필기 ✍️

✏ 예.풀이 답은 p.34

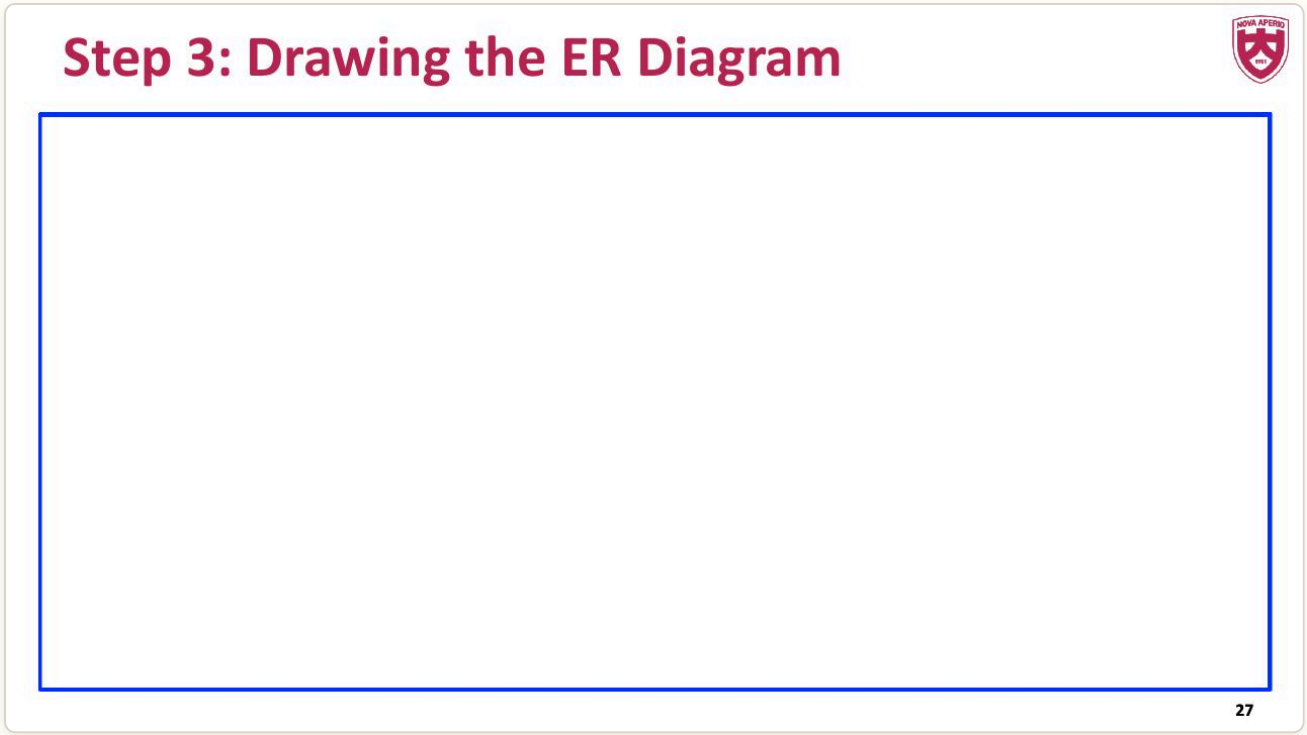


Step 2: REVIEWS (문제)

- REVIEWER와 PAPER를 잇는 관계
  - 관계 속성 3개 — 그중 복합 속성 2개(Comments, Rating)
  - 논문은 심사위원 2~4명이 심사
  - 심사위원은 최소 한 편을 심사해야 기록되고, 여러 편 심사 가능

필기 ✍️

✏ 예.풀이 답은 P.35



Step 3: ER 다이어그램 (문제)

- (빈 그림 — 직접 그려 보기)

필기 ✍️

💡 이해 답 보기 전에 손으로 그려 보고 p.36과 비교



Q&A

- ER Model (Hands-on Practices) — 끝
- p.29~36은 연습 2 정답

필기 ✍️

— 넘어가도 되는 쪽

Step 1: Describing the Entity Types

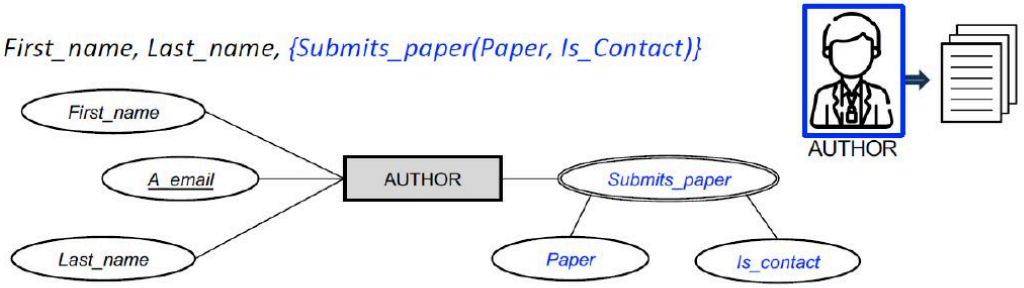
❑ Entity Type 1: AUTHOR

■ Requirements:

- ❑ Authors of papers are uniquely identified by **e-mail id** and have the first and last names
- ❑ Authors can submit one or more papers
  - ■ (Paper) Each can have multiple authors, but one of the authors is designated as the contact author

■ Attributes (List entity types and their attributes based on the requirements):

A_email, First_name, Last_name, {Submits_paper(Paper, Is_Contact)}



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정답 Step 1: AUTHOR

```
A_email, First_name, Last_name, {Submits_paper(Paper, Is_Contact)}
```

- A_email이 키 (밑줄)
- Submits_paper = 다중값 + 복합 (논문, 교신 저자 여부)

필기 ✍️

💡 이해 Submits_paper는 Step 2에서 SUBMITS-IS_CONTACT 관계로 바뀐다

Step 1: Describing the Entity Types (cont.)

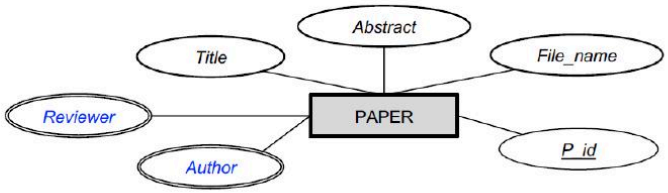
❑ Entity Type 2: PAPER

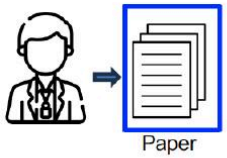
■ Requirements:

- ❑ Each paper is assigned a **unique id** by the system and described by a title, abstract, and the file name
- ❑ Each can have multiple authors, but one of the authors is designated as the contact author
- ❑ A paper is assigned between two and four reviewers

■ Attributes (List entity types and their attributes based on the requirements):

P_id, Title, Abstract, File_name , {Author}, {Reviewer}





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정답 Step 1: PAPER

```
P_id, Title, Abstract, File_name, {Author}, {Reviewer}
```

- P_id가 키, Author·Reviewer는 다중값

필기 ✍️

— 넘어가도 되는 쪽

Step 1: Describing the Entity Types (cont.)

❑ **Entity Type 3: REVIEWER**

■ Requirements:

- ❑ Reviewers of papers are uniquely identified by **e-mail id**
- ❑ Each reviewer has the first and last name, phone number, affiliation, and topics of interest
- ❑ A reviewer rates each paper assigned to her/him on a scale of 1 to 10 in four evaluation categories: (1) technical merit, (2) readability, (3) originality, and (4) relevance to the conference
- ❑ Each reviewer provides an overall recommendation to each paper assigned to her/him
- ❑ Each review contains two types of written comments: (1) one to be seen by the review committee only and (2) the other as feedback to author(s)

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정답 Step 1: REVIEWER — 요구사항

- 이메일로 식별 / 이름·성·전화·소속·관심 주제
- 배정 논문을 4개 항목 1~10점 평가 / 종합 추천 / 심사평 두 종류

필기 ✍️

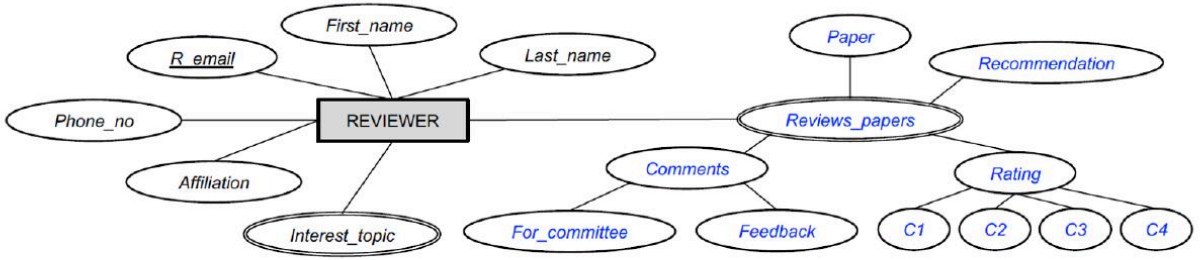
— 넘어가도 되는 쪽

Step 1: Describing the Entity Types (cont.)

❑ Entity Type 3: REVIEWER

■ Attributes (List entity types and their attributes based on the requirements):

R_email, First_name, Last_name, Phone_no, Affiliation, {Interest_topic},
{Reviews_paper(Paper, Recommendation, Comments(Feedback, For_committee),
Rating(C1:Technical_merit, C2:Readability, C3:Originality, C4:Relevance))}



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정답 Step 1: REVIEWER — 속성

```
R_email, First_name, Last_name, Phone_no, Affiliation, {Interest_topic},
{Reviews_paper(Paper, Recommendation, Comments(Feedback, For_committee),
  Rating(C1:Technical_merit, C2:Readability, C3:Originality, C4:Relevance))}
```

- Interest_topic = 다중값 (이중 타원)
- Comments, Rating = 복합 속성

필기 ✍️

★ 시험 Rating(c1~c4), comments(Feedback, For_committee) = 복합 속성 2개

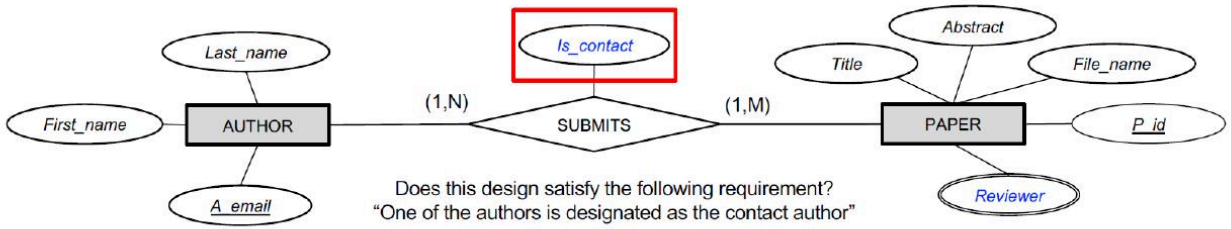
Step 2: Describing the Relationship Types

❑ Relationship Type 1: SUBMITS

■ Relating **AUTHOR** and **PAPER** entity types

- ❑ An author is required to submit a paper and a paper has to be submitted by an author to be recorded
  → both AUTHOR and PAPER types have a total participation (the minimum cardinality = 1)
- ❑ Author can submit many papers and a paper can be submitted by multiple authors (N:M)

■ List and describe relationship types and their cardinality ratios:



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정답 Step 2: SUBMITS

- 저자는 논문을 제출해야, 논문은 저자가 제출해야 기록 → **AUTHOR, PAPER 둘 다 전체 참여** (최소 = 1)
- 저자는 여러 논문, 논문은 여러 저자 → **N:M**
- 그림: AUTHOR (1,N) — SUBMITS — (1,M) PAPER, 관계 속성 Is_contact
  - Q. 이 설계가 '저자 중 한 명이 교신 저자'라는 요구를 만족하나?

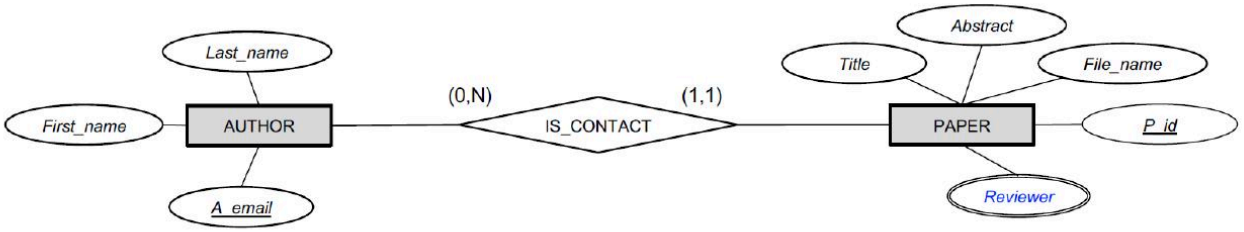
필기 ✍️

⚠ 주의 Is_contact를 SUBMITS 속성으로 두면 '논문마다 정확히 한 명'을 강제 못 함 → 별도 관계 IS_CONTACT (다음 쪽)

Step 2: Describing the Relationship Types

❑ Relationship Type 2: IS_CONTACT

- ■ Relating **AUTHOR** and **PAPER** entity types
  - ❑ For each paper, one author should play the role of "the contact author"
    → PAPER has a total participation (the minimum cardinality = 1)
  - ❑ An author can be the contact for many papers but does not have to be the contact for any papers
    → AUTHOR has a partial participation (the minimum cardinality = 0)
- ■ List and describe relationship types and their cardinality ratios:



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정답 Step 2: IS_CONTACT

- 논문마다 저자 한 명이 교신 저자 → **PAPER 전체 참여** (최소 = 1)
- 저자는 여러 논문의 교신 저자일 수 있지만 아니어도 됨 → **AUTHOR 부분 참여** (최소 = 0)
- 그림: AUTHOR **(0,N)** — IS_CONTACT — **(1,1)** PAPER

필기 ✍️

★ 시험 PAPER (1,1) = 교신 저자는 딱 한 명

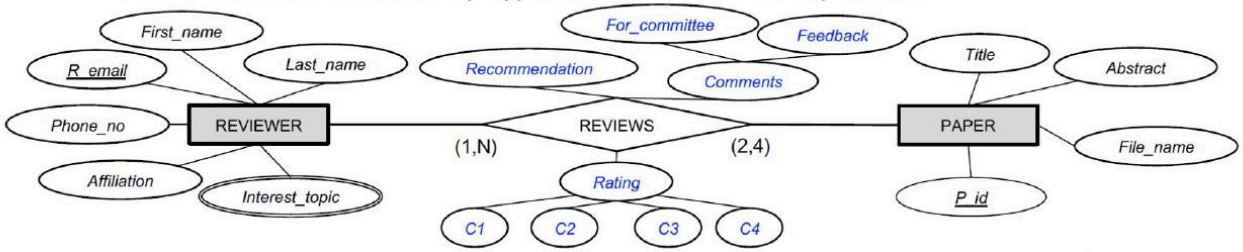
Step 2: Describing the Relationship Types

❑ Relationship Type 3: REVIEWS

■ Relating **REVIEWER** and **PAPER** entity types

- ❑ It has three relationship attributes including two composite attributes (*Comments* and *Rating*)
- ❑ A paper is reviewed by 2 to 4 reviewers
  → its (min, max) constraint is (2,4) (i.e., total participation)
- ❑ Every reviewer has to review at least one paper to be recorded and can review many papers
  → its (min, max) constraint is (1,N) (i.e., total participation)

■ List and describe relationship types and their cardinality ratios:

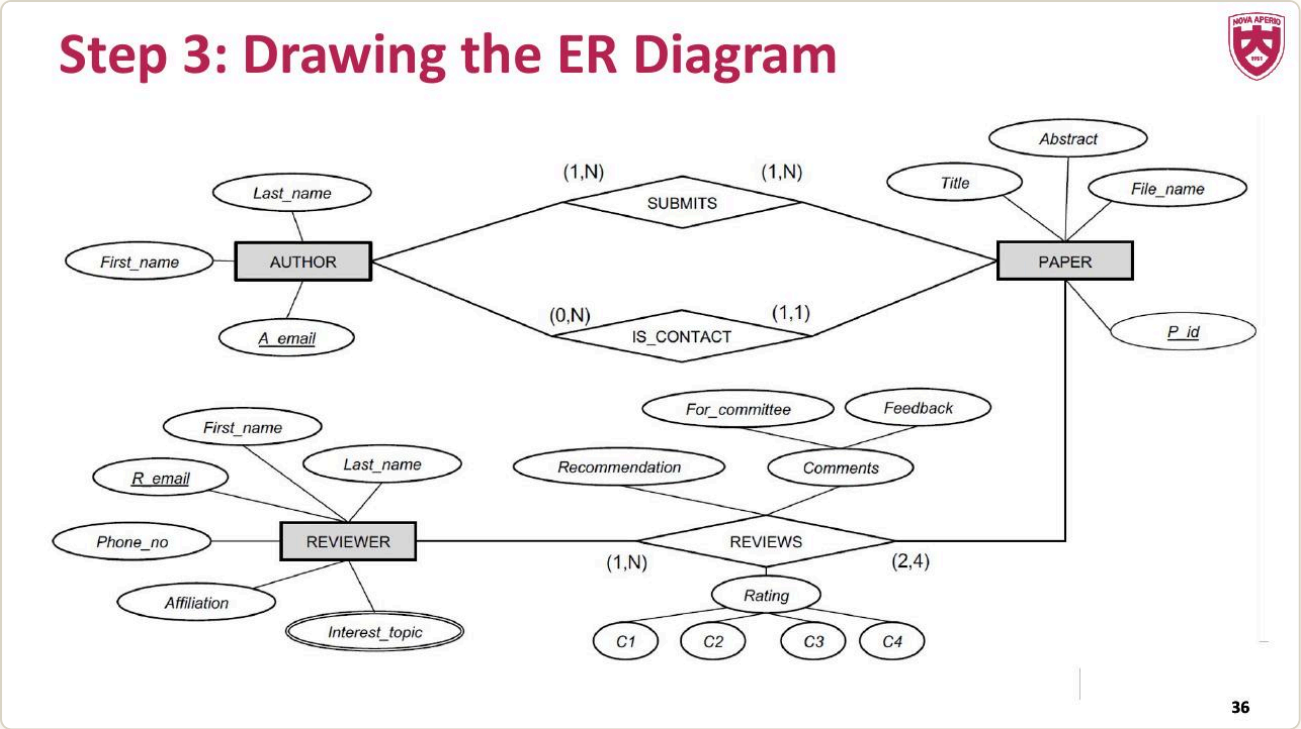


정답 Step 2: REVIEWS

- 관계 속성 3개 — 복합 2개(Comments, Rating) + Recommendation
- 논문은 심사위원 2~4명 → PAPER **(2,4)** (전체 참여)
- 심사위원은 최소 한 편, 여러 편 가능 → REVIEWER **(1,N)** (전체 참여)
- 그림: REVIEWER (1,N) — REVIEWS — (2,4) PAPER, Rating(C1~C4), Comments(For_committee, Feedback)

필기 ✍️

★ 시험 (min,max)는 그 엔티티 입: '논문은 2~4명'
⟶ PAPER 쪽 (2,4)



Step 3: 완성 ER 다이어그램 (CONFERENCE)

- AUTHOR(A_email, First_name, Last_name)
- PAPER(P_id, Title, Abstract, File_name)
- REVIEWER(R_email, First_name, Last_name, Phone_no, Affiliation, {Interest_topic})
- AUTHOR (1,N) — **SUBMITS** — (1,N) PAPER
- AUTHOR (0,N) — **IS_CONTACT** — (1,1) PAPER
- REVIEWER (1,N) — **REVIEWS** — (2,4) PAPER, 관계 속성 Recommendation, Comments(...), Rating(C1~C4)

필기 ✍️

★ 시험 정답 요약: SUBMITS M:N 양쪽 total · IS_CONTACT AUTHOR (0,N)·PAPER (1,1) · REVIEWS REVIEWER (1,N)·PAPER (2,4)